

CS4-Materials Selection for a Ferritic / Austenitic Stainless Transition Weld with a proposal for a Non-Destructive Testing Method to ensure Volumetric Weld Integrity

A dissimilar transition weld between ferritic steel and austenitic stainless steel is considered for a high pressure corrosive system. Joining **two different metals** by **welding** them together.

- One is a **ferritic steel** (SA508) → strong and tough
- The other is an **austenitic stainless steel** (305) → corrosion resistant
- **filler metal** (309) → a material that melts between the two and helps form the weld

Problem: These metals are **very different** in chemical composition and crystal structure:

- SA508 is **ferritic** (wants to form martensite if cooled too fast = brittle)
- 305 is **austenitic** (good corrosion resistance, but soft)
- Mixing them during welding can result in **bad structures**.

Goal:create a weld zone that is resistant to corrosion,wont crack and can withstand high pressures.Then use NDE to evaluate the weld zone.

This is controlled by choosing the right filler material;this controls how much of each parent mixes into the weld=dilution.

Dilution factor controls chemical composition → which controls microstructure → which affects weld strength, corrosion resistance, and cracking risk.-want a dilution factor with a ferrite number of 10.

WELDING

MATERIAL PROPERTIES -chemical composition means likely to be austenitic

FERRITIC: needs to withstand high pressure forces (strength) and absorb lots of energy before fracture (toughness)

AUSTENITE: needs to withstand highly corrosive environments (corrosion resistance)

Question 1:**What are the material properties of a filler?**

WELD FILLER: should be at least equal in strength and corrosion properties to the poorest component in the joint to prevent it from becoming a critical weak point. **Select filler metal that is appropriate for the welding process you select with a nominal UTS of >500MPa.**

MICROSTRUCTURE: The microstructure must be resistant to cracking and a measure of a property known as the Ferrite Number (FN) needs to be considered in relation to this.

FERRITE NUMBER (FN)=amount of ferrite,a crystalline structure,in a weld metal.

The chemistry of the weld filler and associated parent material(s) can be determined by using FN.

MATERIAL SELECTION

Parent material(s) selection:

• Select a ferritic carbon alloy with:

- Yield > 550MPa
- UTS = 650 – 950MPa
- Elongation >15%

Select an austenitic stainless steel with:

- Yield > 200MPa
- UTS > 550MPa
- Elongation >25%

• Select filler metal that is appropriate for the welding process you select with a nominal UTS of >500MPa.

Weld is between a low carbon alloy steel (SA508) and an austenitic stainless steel (305) using an austenitic weld filler (309)

Material	C	Si	Mn	Ni	Cr	Mo	Nb	Cr Equiv	Ni Equiv
SA 508	0.19	0.1	0.25	3.4	1.6	0.45	0.01	2	9
Type 305	0.05	0.25	1.0	10	19.0	0.05	0.05	19	12
Type 309	0.01	0.65	1.0	12	25.0	0.25	0.01	26	13

CHEMICAL COMPOSITION

Carbon: Precipitation and solid solution strengthening.

Silicon: Used for steelmaking purposes; killing the steel and removing oxygen. i.e improves resistance to oxidation. (important for high integrity)

Manganese: Improves strength and hardness

Chromium: Provides corrosion resistance and strength

Nickel: improve toughness and hardenability(flexible, tough, less likely to crack)

Molybdenum: Enhances creep resistance

Niobium: Grain refinement and precipitation hardening

NICKEL AND CHROMIUM EQUIVALENTS

Both chromium and nickel are used in steel alloys for physical and chemical benefits. use these equivalents to create a strong weld.

The nickel equivalent and chromium equivalent differ from the nickel content and chromium content, respectively.

$$\text{Cr Equiv.} = \text{Cr} + \text{Mo} + 1.5\text{Si} + 0.5\text{Ni}$$

$$\text{Ni Equiv.} = \text{Ni} + 30\text{C} + 0.5\text{Mn}$$

Higher chromium equivalent will promote ferritic microstructure as its composed of ferritic stabilisers, likewise higher nickel equivalent will promote austenitic structure as its composed of austenite stabilisers.

WELDING CONSIDERATIONS

Dilution control: the resultant chemical composition of the fusion zone is related to the degree of mixing between the filler metal and the two parent metals.

Sensitisation: where carbides form from chromium, and essentially deplete the chromium content.

Unwanted phases: if metal is unstable, delta ferrite and sigma ferrite phases can form and degrade to brittle intermetallic phases, becoming susceptible to corrosion and fracture.

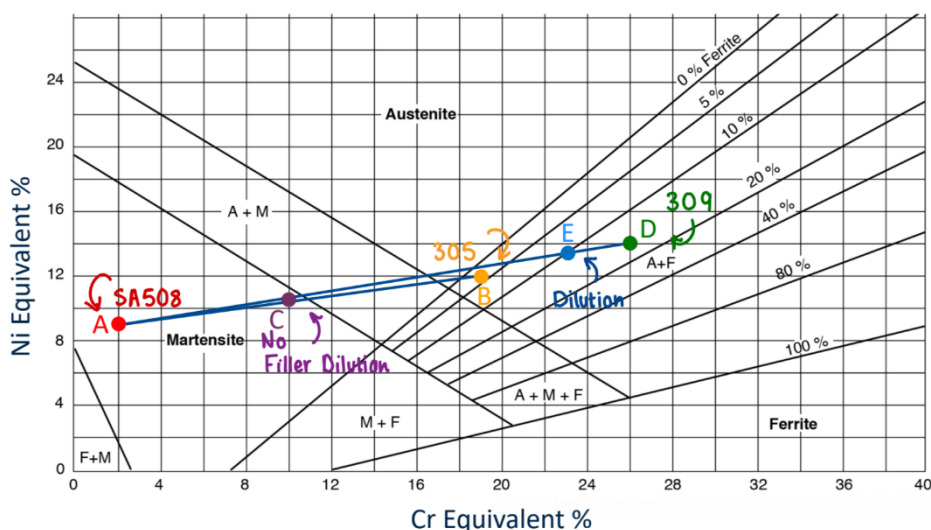
Ferrite Number (FN): A low ferrite number suggests larger proportion of austenite and hence better corrosion resistance, while a high ferrite number infers higher cracking resistance. **Ideal FN for dilution of weld filler in this case = 10, which gives good balance of crack resistance and corrosion properties.**

How to use the Schaeffler diagram:

graphical tool used to predict the microstructure of stainless steel welds by plotting the chromium and nickel equivalents of the base metal and filler metal. you use **known or assumed dilution ratios** to determine where to plot your final weld composition **on the diagram**.

Displacement of a point **on the diagram** shows how the **Cr/Ni balance shifts** as dilution changes — it's a **visual representation** of composition change. We want ferrite number 10—that's where point E is. Point E is then the target composition.

SCHAEFFLER DIAGRAM = filler selection process.



- Point A: SA508 Ferritic
- Point B: 305 Austenite
- Point C: Welded area if no filler weld was used
- Point D: 309 Filler
- Point E: Target composition (FN=10) of weld controlled by dilution factor of the two parent metals and weld filler.

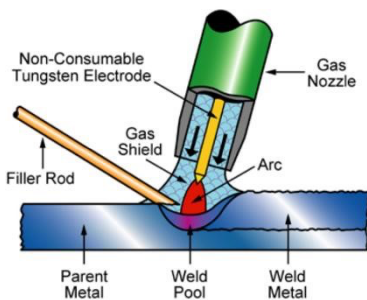
Position (and hence composition) of E also depends on the size of the weld. A wider weld means a larger dilution factor and hence larger displacement on Schaeffler diagram.

Point C has a FN of 0 and will therefore have very weak crack resistance due to martensite formation. • This is why an appropriate filler is used, Point D. • Dilution of the weld filler with the parent material can be controlled to create Point E. In this example the weld with composition Point E would be ideal with FN = 10-11 (acceptable range = 5 – 20). This gives a good balance of crack resistance and corrosion properties.

- Displacement of E is determined by the dilution factor; this is mostly determined by the geometry of the weld. In reality the weld should be very very small, but we assume 25-30% dilution anyways.
- A **larger weld pool** melts more base metal → **higher dilution**
- A **narrow weld bead** (e.g., low heat input) melts less base metal → **lower dilution**

So, **bigger welds = more displacement** away from the filler's position.

WELDING PROCESS



The weld volume in this context is very small with steep fusion angles of about 4 degrees, resulting in very tight join gaps. Hence there isn't much of a dilution factor.

TIG (Tungsten Inert Gas) performs well in this kind of weld design as it is well suited for narrow joints and steep angles with fine control of heat input. Inert gas also prevents oxidation

NON-DESTRUCTIVE EVALUATION (NDE)

NDE is comprised of seven main areas, two of which (highlighted) are the main focus for this study: Flaw detection and evaluation and Location determination and evaluation.

NDE METHODS: RADIOGRAPHIC VS ULTRASONIC TESTING

NDE method needs to be a volumetric inspection method that is able to both locate and size the defects;

There are 2 volumetric NDE methods.

1. Radiographic Testing (RT)
2. Ultrasonic Testing (UT).

- UT is superior to RT in detecting surface defects, such as a toe crack.

UT is very versatile it is able to size and locate defects, whereas RT don't offer the same resolution in depth due to the nature of radiography and hence locating defects is more difficult using RT.

TYPES OF ULTRASONIC TESTING:

UT is a method capable of detecting defects of variable orientations in terms of tilt angle and size.

- UT utilises the propagation of sound waves in the object or material tested.

1. Longitudinal waves
2. Transverse Waves

Longitudinal waves: produced by voltage applied to a piezo-electric transducer.

Transverse (shear) waves: require more acoustically solid material for effective propagation (not effective in liquids and gases).

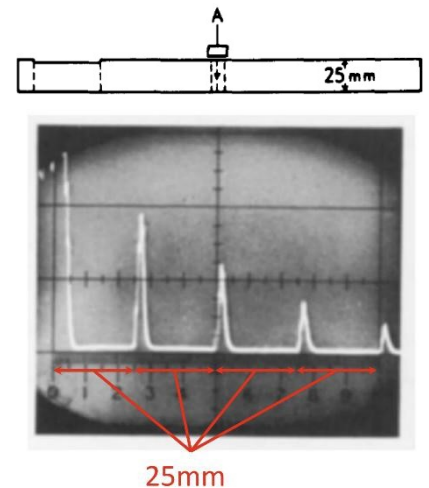
Longitudinal waves are approx. twice the speed of shear waves, hence wavelength (resolution) of transverse waves is roughly half, making it more sensitive to smaller defects.

However, propagation of transverse waves is more difficult through anisotropic structure of weld. So ultimately, sizing tolerance is hindered because longitudinal waves (which suffer from worse resolution due to larger wavelength) are more effective.

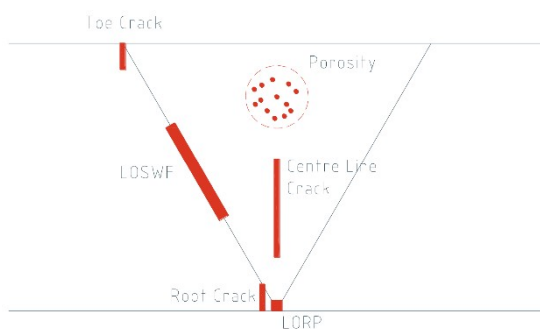
HOW DOES UT WORK:

Signals representing interface reflection are produced: Initial bang followed by evenly spaced repeating backwalls (intervals depending on component thickness) of decreasing amplitude.

If there is a defect, the waves will reflect in between these fixed intervals. This will appear as a smaller sharper new echo (peak) before the first backwall echo depending on the defect size/orientation.



DEFECT TYPES & DETECTION METHOD:



Toe Crack: two techniques could be used:

- Transverse waves from a 45° probe
- Creep waves (78° longitudinal surface waves) LOSWF: best response given by 60° transverse probe which will strike defect at 90°
- LORP: 45°, 60° or 45° transverse probes from either side of centreline

Root Crack: similar to LORP, but will not produce defined rise/fall signals and instead will produce multiple peaks and troughs.

Porosity: Difficult with UT as signals produced would be very weak, RT can be more effective

Centreline Crack: uses a transmitting shear probe and a receiving shear probe in a tandem setup.

Defect Type	Detection Method
Toe Crack	UT; transverse waves at 45°, or creep waves
LOSWF	UT; transverse waves at 60°
LORP	UT; transverse probes preferred at 45°
Root Crack	UT; transverse probes preferred at 45°
Porosity	Difficult with UT; RT can be more effective
Centreline Crack	UT; using two transverse probes

MISCELLANEOUS – QUESTIONS & CONSIDERATIONS

How do we filter out more esoteric alloys in material selection?

Exotic materials can have excellent properties which are far in excess of what's required -- most popular ones tend to be the cheaper ones, which we can manufacture in bulk. They need to be produce to a large scale and sufficient volume, hence cost becomes a factor that cannot be ignored

Material index considerations

We need to maximise both fracture toughness and yield strength, not necessarily a ratio between them. Density is not an important factor for this context.

